CSD361 - Lab Assignment 1

Data Preprocessing and Linear Regression

Deadline - 22 September 2026, 11:59 PM.

Objective

Develop a **Linear Regression model to predict taxi trip fare** and investigate how different normalization techniques affect model training, parameter estimation, convergence, and prediction performance.

Submission Guidelines:

1. **Read each question carefully** and provide a clear, well-structured solution.
2. For programming or implementation-based questions :
    - Write readable and well-commented code, and its output screenshot is required.
    - Clearly mention the dataset, preprocessing steps, algorithms, and parameters used.
    - Include relevant output, graphs, or performance measures where required.
3. Prepare the documentation with all code screenshots and output screenshots (wherever applicable) and upload a PDF.
4. Also, provide the link to your code file in the document itself (for example, Jupyter Notebook, lab, or Colab), whatever you are using. We will be using the same code file during the viva for this assignment.
5. This exercise depends on several Python libraries; load the required libraries to help with data manipulation, machine learning tasks, and data visualization.
6. Download the dataset here if you are unable to Chicago Taxi Trips Dataset
7. The deadline for the submission is 22 September 2026, 11:59 PM.
8. **Please note that this assignment contains 50 marks, and 50 marks will be for the related VIVA. Total -100 marks.**

I. Dataset Exploration

Download the Chicago Taxi Trips dataset (given above in the submission guidelines).

TASKS:

1. Determine the number of observations and attributes.
2. Identify:
    - Numerical attributes
    - Categorical attributes
    - Identifier attributes
    - Timestamp attributes
3. Identify missing values in each attribute.
4. Identify duplicate records.
5. Calculate descriptive statistics for:
    - trip_seconds
    - trip_miles
    - fare
    - tips
    - tolls
    - extras
6. Plot the distributions of the numerical variables.

II. The Prediction Problem - Linear Regression

TASKS:

1. Develop a Linear Regression model to predict FARE using appropriate features from the dataset.
2. Provide Justification for the features you choose to develop a model.

HINT: You may consider: `trip_seconds, trip_miles,` pickup/drop-off information, temporal features extracted from timestamps, other features that are logically available at prediction time

III. Build Three Regression Models

TASKS:

1. Construct the following models:
    a. Model 1 – Original Features - Train Linear Regression using the original numerical features.
    b. Model 2 – Min-Max Normalized Features - Train Linear Regression after Min-Max normalization.
    c. Model 3 – Standardized Features - Train Linear Regression after Z-score standardization.

2. Compute MAE and MSE loss. Fill in the table with the values you obtain:

Model	MAE	MSE
Original		
Min-Max		
Standardized		

3. Does normalization change the obtained MAE or MSE?
4. Does the prediction performance remain almost identical for all the 3 models? Why?
5. Which model would you recommend and why?

IV. Investigate the Effect of Scaling on Gradient Descent

Use the model given in III (given above)

TASKS:

1. Use the same learning rate initially. Plot: MSE vs. Iteration for all three models.
2. Using the above graph, compare all the 3 models using the
    a. Number of iterations required for convergence.
    b. Final MSE.
    c. Sensitivity to the learning rate.
3. Experiment with the following learning rates: 0.00001, 0.0001, 0.001, 0.01, 0.1, and determine which learning rates lead to: (you may justify your answer using plots as well)
    a. Convergence
    b. Slow convergence
    c. Divergence
4. Explain why normalization changes the behavior of Gradient Descent.

V. How Do Outliers Affect Normalization?

TASKS:

1. Identify potential outliers in trip_seconds, trip_miles, and fare.
2. Show the learned model on Dataset A - Original data, Dataset B - Data after removing extreme outliers by training Linear Regression on both datasets.
3. Does removing outliers necessarily improve Linear Regression performance? Justify your answer using your experimental results.

**********End of the Assignment**********